

IOE 534 Project #2: Biomechanics Modeling Using 3DSSPP and OpenSim/OpenCap

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Squat & Stoop Motion

Anthropometric data

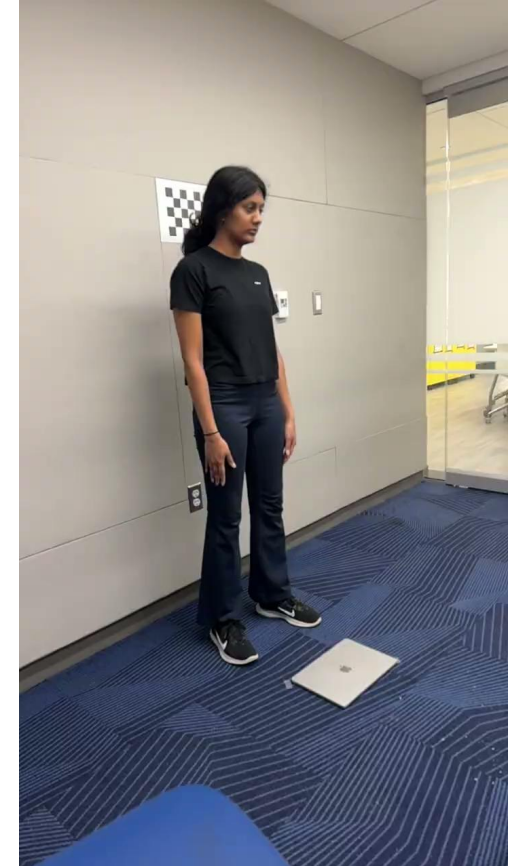
- Female
- Wt(Kg) 57.0
- Ht(cm) 162.0

Weight of Laptop

- 1.24 kg



Squat lifting motion



Stoop lifting motion

Squat vs. Stoop Lifting: Quantitative Biomechanical Comparison

Variable	Squat Lifting	Stoop Lifting
Trunk angle (Kinematics)	More upright trunk	Greater trunk flexion
Knee flexion	High	Low
Hip flexion	Moderate	High
Lumbar ROM	Lower	Higher Lumbar flexion
Lumbar compressive forces (Kinetics)	Similar or slightly ↑	Similar
Shear forces	Lower	Slightly ↑
Muscle demand	↑ Quadriceps	↑ Back extensors
Energy cost / fatigue	Higher	Lower
Spinal stability	More muscular support	More passive tissue reliance
Natural preference	Less commonly used alone	Often part of hybrid strategy

3DSSPP Models

Squat

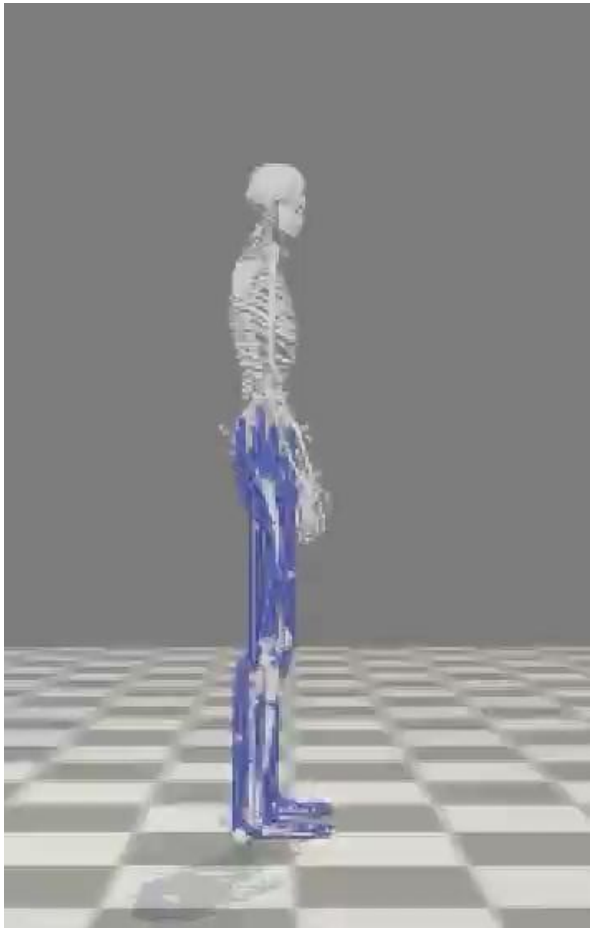


Stoop

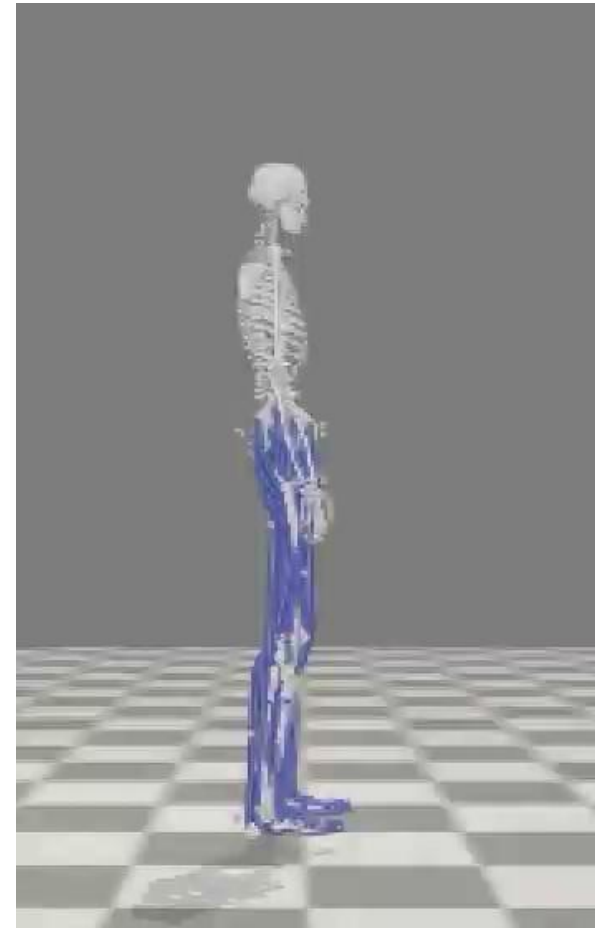


OpenSim Models

Squat



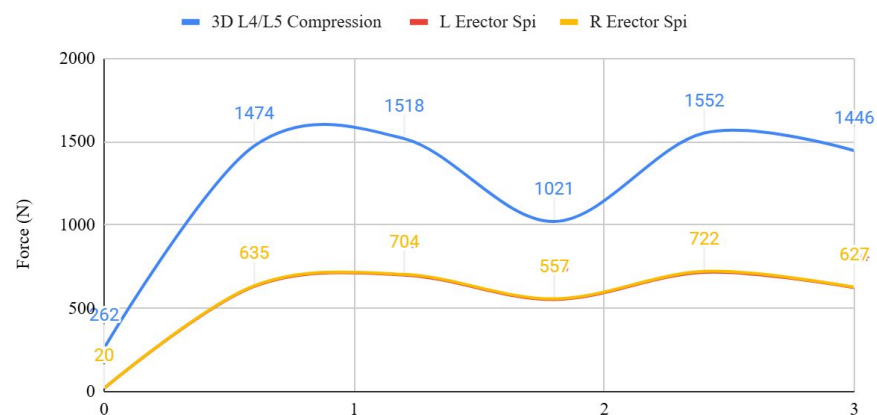
Stoop



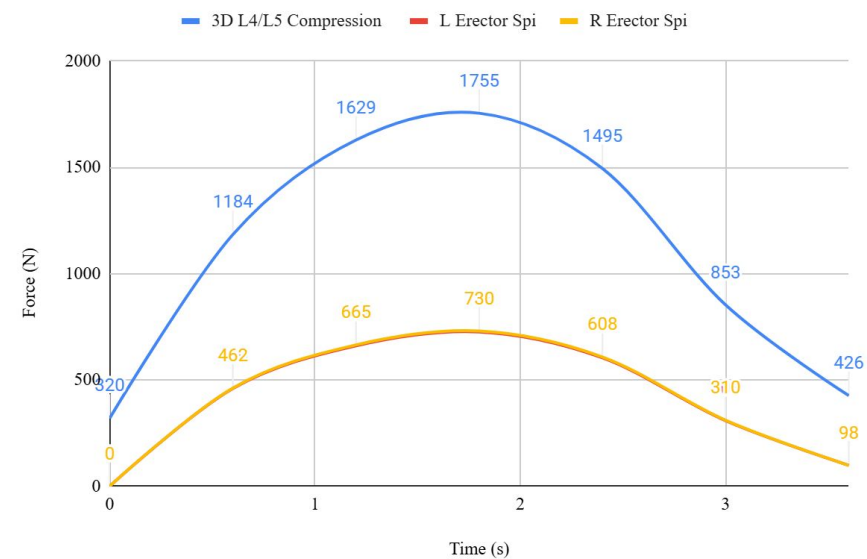
Estimation of Maximal Lumbar Forces

3DSSPP

Stoop Lumbar Force

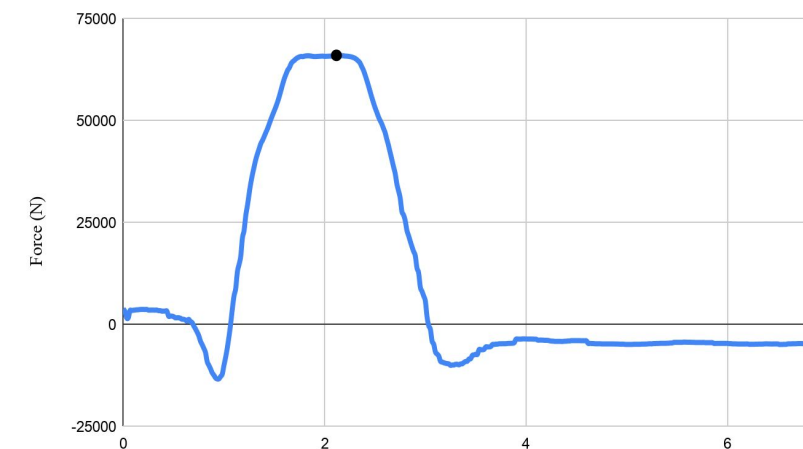


Squat Lumbar Force

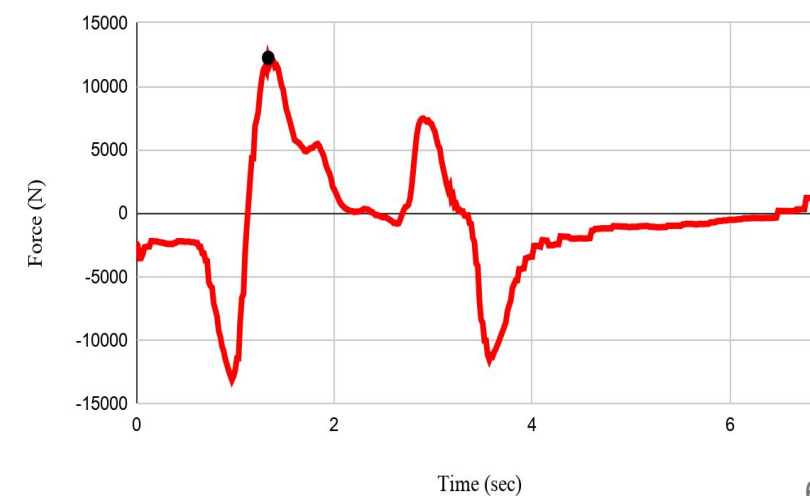


OpenSim

Back-on-Torso Force (Fy) vs. Time during Stoop Lift



Back-on-Torso Force (Fy) vs. Time during Squat Lift



OpenSim vs 3DSSPP: Back Load Differences



Dynamic & Time-based

vs

Static & Posture-based

Joint reaction force
(global frame)

vs

L4/L5 compression
(disc direction)

Joint reaction–based loading

$\approx$ External forces
+ Back joint reaction force

vs

More complete spinal loading

$\approx$ Joint loading & equilibrium moments
+ Optimized trunk muscle forces
+ Abdominal pressure

They represent different biomechanical quantities

Sources of Differences of Our Model

The quantity most comparable to 3DSSPP L4/L5 low-back compression can be conceptually approximated as:

$$F_{\text{comp,approx}} \approx -\mathbf{F}_{JR} \cdot \hat{\mathbf{n}}_{\text{disc}} + \sum_i \max(0, \mathbf{F}_{\text{muscle},i} \cdot \hat{\mathbf{n}}_{\text{disc}}) - F_{\text{abd}}$$

F_JR = lumbar/back joint reaction force

F_muscle = muscle forces

F_abd = abdominal pressure

n_disc = local disc normal or spinal axial unit vector

However, in the current OpenSim workflow, **only the first term is available in practice**, namely the joint reaction force contribution.

Parametric Study

Trial design:

One-factor-at-a-time analysis for stoop posture was used.

For each trial, only one variable was changed, while all others were fixed at preset reference values:

- **23 kg** hand load
- **25 cm** horizontal distance
- **0 cm/75 cm** vertical distance
- **0°** asymmetry angle

Parameter	Value	Max Lumbar Force (N)	L erector Spi (N)	R erector Spi (N)
Hand load	15 kg	2586	1177	1183
	23 kg	3416	1573	1582
	30 kg	3606	1645	1654
Horizontal Distance	20cm	3167	1444	1453
	25cm	3416	1573	1582
	30cm	3848	1782	1792
Vertical Distance	0cm	3416	1573	1582
	50cm	2945	1282	1289
	100cm	2894	1244	1251
	175cm	2746	1171	1178
Asymmetry Angle	0°	1884	722	727
	45°	2325	276	1361
	90°	3698	0	1702

Parametric Study 1 - Hand Loading

Loading levels:

1. 15 kg
2. 23 kg
3. 30 kg

*(gravity=9.8 m/s²)



Parametric Study 2 - Horizontal Location of the Hands

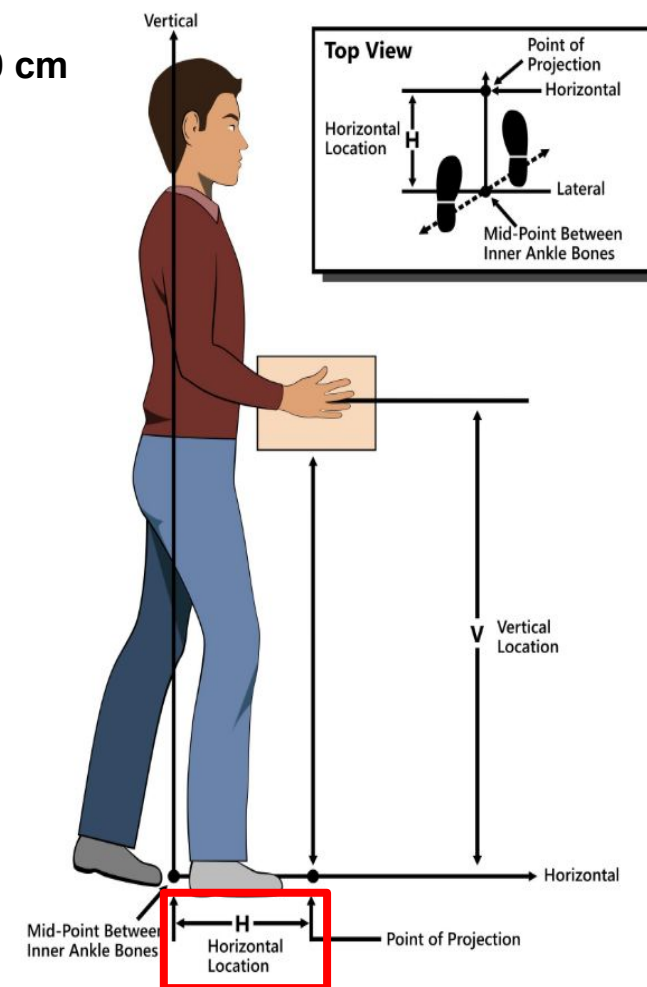
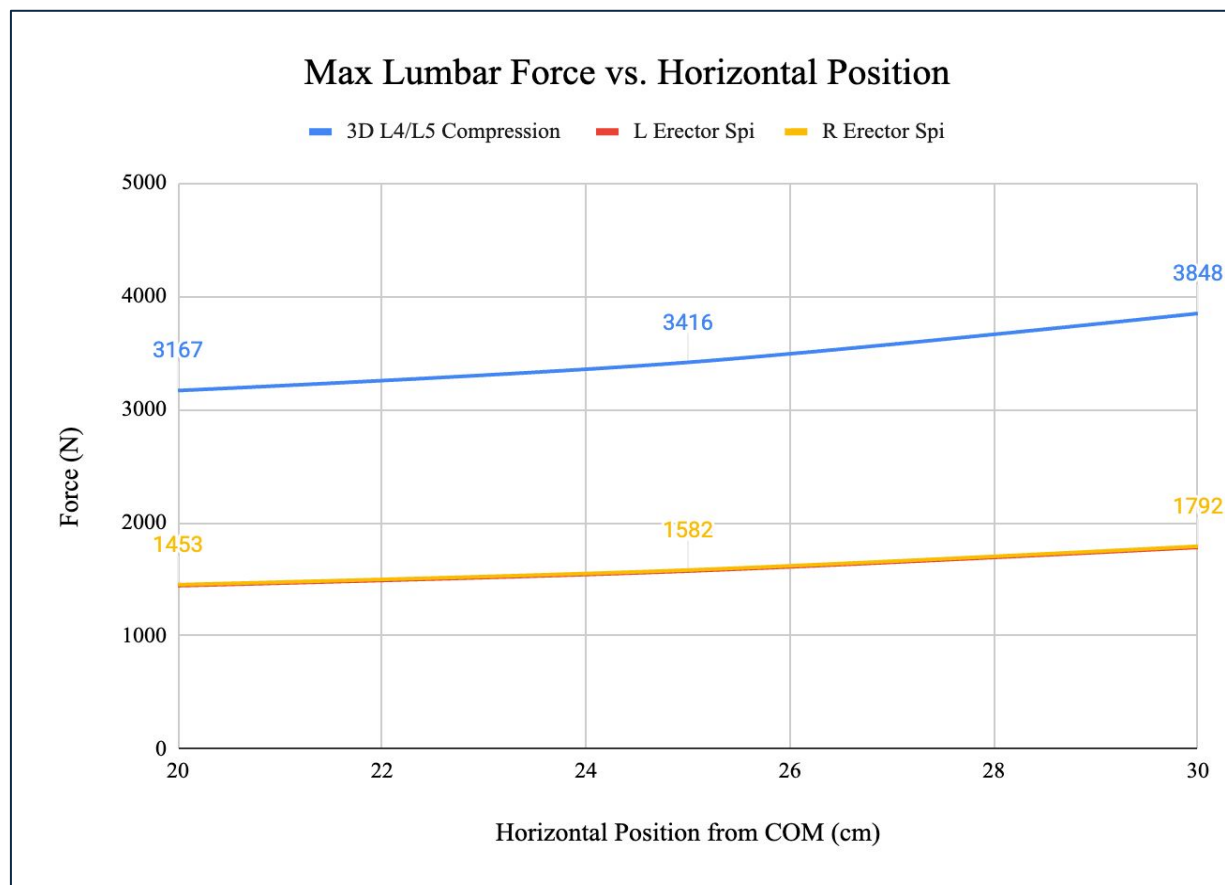
Level:



Minimum = 0 cm

RNLE Value : 25 cm

Maximum = 30 cm



Parametric Study 3 - Vertical Location of the Hands

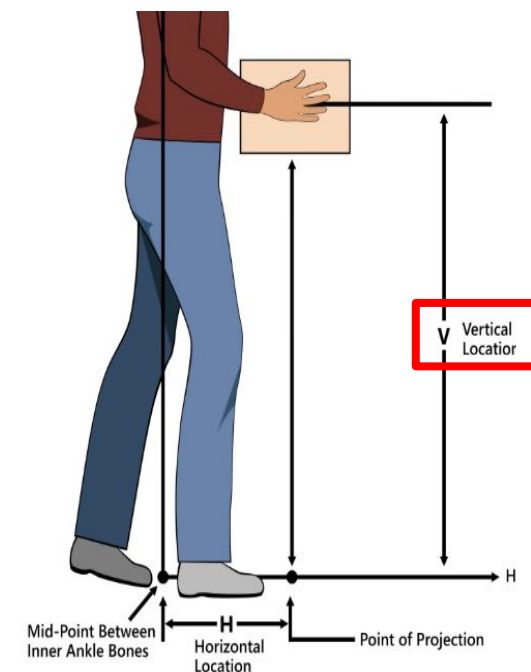
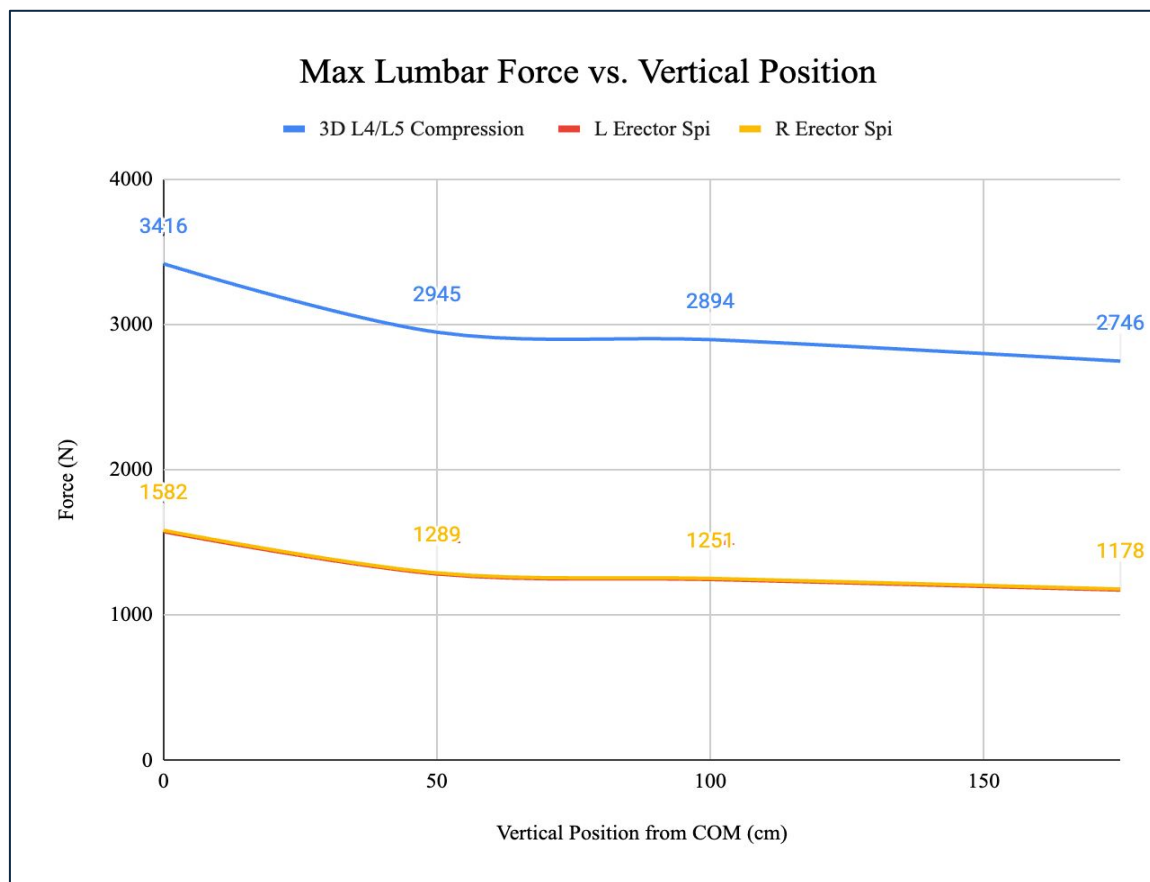
Level:



Minimum = 0 cm (floor level)

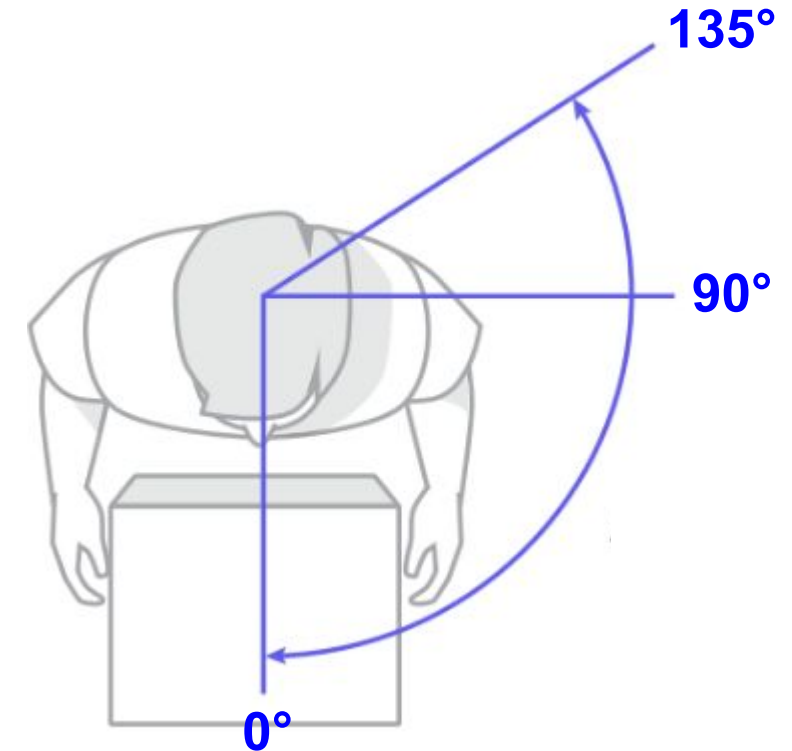
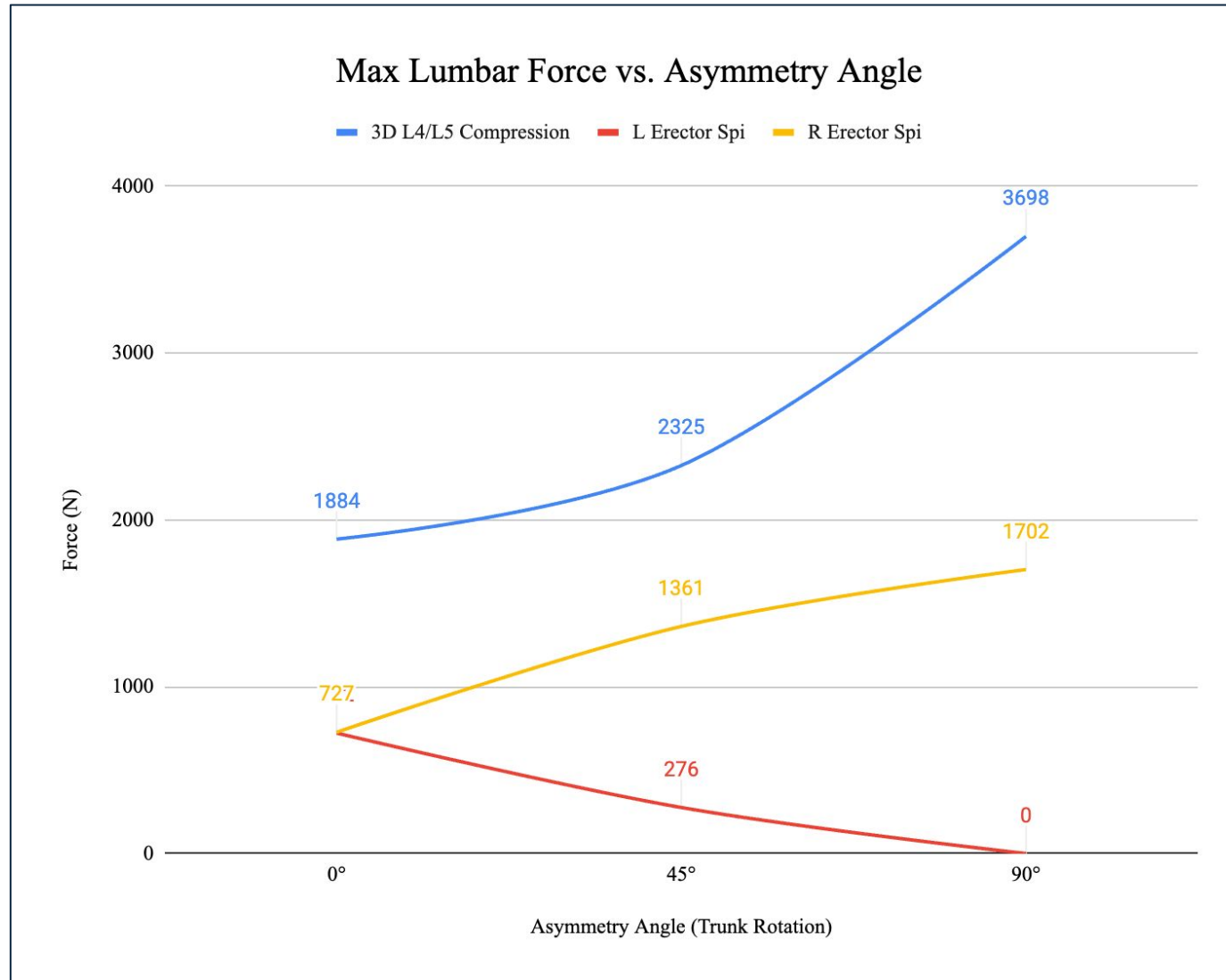
RNLE Value : 75 cm

Maximum = 175 cm

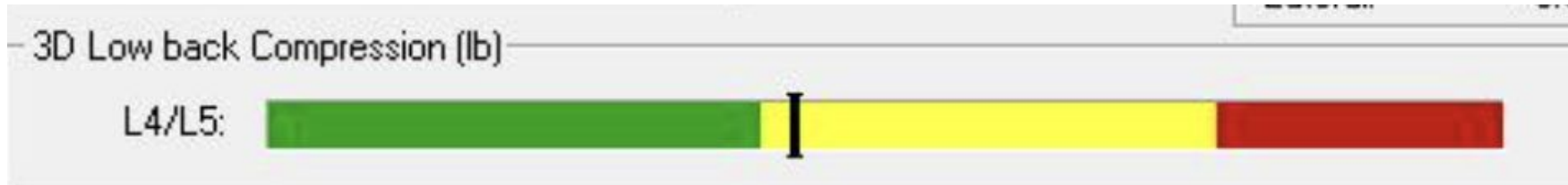


Parametric Study 4 - Asymmetry Angle

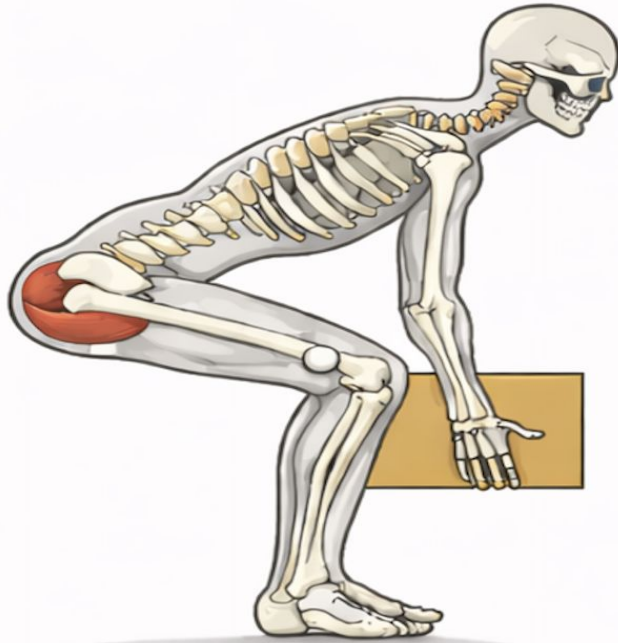
Asymmetry Angle: how far the object is displaced from the front of the body



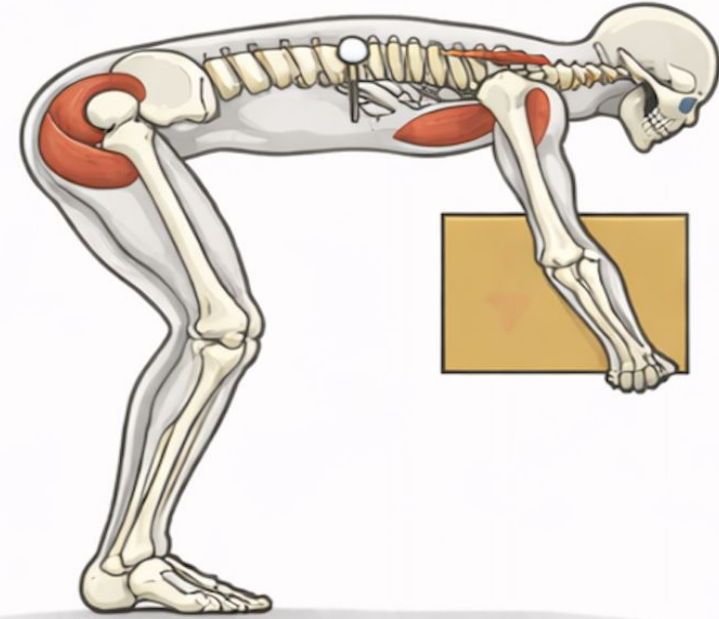
Limitation



- Even when lumbar compressive forces remain below the 3.4 kN threshold, injury risk is not eliminated.
- Repetitive loading over time can lead to cumulative damage.
- A key limitation of both 3DSSPP and OpenSim is that they do not model muscle fatigue, which plays a critical role in real-world injury development.



Squat Lifting



Stoop Lifting

Safe Lifting Recommendations

- **Differences between lifting styles were relatively small**
- **No single lifting technique is always safest**
- **Keep loads ≤ 23 kg (51 lb) under ideal conditions**
- **Hold the load close to your body (≤ 25 cm)**
- **Lift at waist height (~ 75 cm) when possible**
- **Avoid twisting** → keep your body straight
- **Use a good grip or handles**
- **Reduce how often and how far you lift.**
- **Lifting speed affected results** (slower lifting → lower loads)